AI-DRIVEN ACADEMIC RISK & DROPOUT EARLY-WARNING SYSTEM USING DIGITAL TWIN

A PROJECT REPORT

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CO-PO, PSO AND SDG GOALS MAPPING

CO1: Identify any real-world problem and find the optimal solution by applying gathered knowledge and skills.

Course outcomes	POs											PSOs			SDGs
CO1	1	2	3	4	5	6	7	8	9	10	11	1	2	3	4, 9, 10
	3	3	3	-	3	-	2	-	-	-	-	3	3	-	

Mapping Justification

CO-PO Mapped	Justification
CO1 – PO2	Analyses student records to identify academic risks.
CO1 – PO3	Develops a system for early dropout-risk detection.
CO1 – PO5	Uses modern tools for student data analysis and risk prediction.
CO1 – PO6	Addresses student dropout through timely academic support.
CO1 – PO8	Ensures student privacy and ethical data handling.

CO-PSO Mapped	Justification
CO1 – PSO1	Applies programming skills to develop the student monitoring system.
CO1 – PSO2	Uses modelling and prediction to identify students at academic risk.

SDG	Justification
4 – Quality Education	Supports early identification of academic difficulties and timely student guidance.
9 – Industry, Innovation and Infrastructure	Uses AI and digital technology to improve academic monitoring.
10 – Reduced Inequalities	Promotes inclusive student support through early risk detection and personalized intervention.

ABSTRACT

In Colleges, identifying an underperforming undergraduate is often too late, since we only know about them once end-semester grades are declared. Currently, records are maintained in different forms such as paper-based attendance registers, standalone spreadsheets, and separate departmental dashboards. We developed and implemented Digital Class Twin, a web application that monitors student performance on a daily basis and notifies mentors before declining attendance and grades become permanent. Our backend is developed using FastAPI, SQLAlchemy and SQLite, while our frontend is a modern and interactive React 18 based application. Digital Class Twin stores the daily lectures attended and internal assessment marks, and categorizes each student into one of three basic and easy-to-understand states: Stable, Warning, or Critical. To prevent false alarms due to temporary fluctuations in scores, we have added a hysteresis margin of 3.0 points in our state machine. When a student's status changes, an audit log entry is created and customised messages are sent out to the mentor, parent and student. Digital Class Twin has separate pages for the student, mentor and supervisor. Alongside showing attendance charts and warnings in real time, the application also features a live What-If simulator, allowing students and mentors to simulate scenarios, such as calculating how many more classes a student needs to attend to comfortably clear the university 75% attendance requirement.

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LIST OF SYMBOLS ABBREVIATIONS

Term / Symbol	Definition / Description
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
Bcrypt	Cryptographic Password Hashing Algorithm
CGPA	Cumulative Grade Point Average
CIA	Continuous Internal Assessment
CORS	Cross-Origin Resource Sharing
CPU	Central Processing Unit
EDM	Educational Data Mining
EWS	Early Warning System
FR	Functional Requirement
GPA	Grade Point Average
JSON	JavaScript Object Notation
JWT	JSON Web Token
REST	Representational State Transfer
SHAP	Shapley Additive Explanations
SQLite	Self-contained Lightweight SQL Database Engine
XAI	Explainable Artificial Intelligence
H_composite	Composite Student Health Score
R_risk	Dropout Risk Index
S_acad	Normalized academic performance score
S_att	Rolling 30-day class attendance percentage
S_beh	Normalized behavioral conduct index (0 - 100)
V_score	Academic grade volatility metric indicating score instability

CHAPTER 1

INTRODUCTION

1.1 Basics of Digital Class Twin and Educational Analytics

Digital Class Twin brings the core idea of twin modeling into an engineering college. Of checking students only once a term with static transcripts Digital Class Twin records daily classroom data. This includes attendance for each period Continuous Internal Assessment marks and lab submissions. By doing Digital Class Twin keeps an up-to-date digital profile for every learner. Educational Analytics acts as the mind behind this system.

Educational Analytics acts as the brain of this setup. Educational Analytics helps advisors spot downward trends long before formal examinations take place.

1.1.1 Introduction

During a semester, a student's academic performance does not always follow a steady flow. Students may be absent for a few classes, score less marks in class tests or become late to submit their assignments. When these issues are occurring rarely, they do no have a major impact on the students academic performance. But when these issues occur regularly across many subjects, the students academic performance becomes serious before faculty understood the problem.

According to Anna University, each student must maintain at least 75% attendance in a semester in order to become eligible for end semester examinations. So identifying these issues early provide faculty sufficient time to provide remedial classes and guide students.

1.1.2 Existing System and Its Limitations

In most colleges, academic profile of a student is maintained in different files rather than in one centralized system. Faculty generally mark attendance, internal marks and assignment marks in different modules in a software. Because of this, faculty do not get a complete status of a student's academic status across the semester.

As a result, reviewing a student's academic performance are conducted once or twice during a semester, by the time issues are identified, it may be difficult for the students to improve and other supporting activities such as remedial classes.

1.1.3 Digital Class Twin Concept

The proposed system uses the concept of digital twin technology, which is now commonly used in fields like aerospace engineering and manufacturing to monitor their equipments and identify issues at an early stage. We use this concept to monitor and understand student's academic status.

In this system, each student's digital twin combines several records such as attendance, continuous assessment test(CAT) marks, laboratory marks, assignment marks and disciplinary records. Whenever a new data like attendance or marks are entered, the system updates the profile and provide current risk level. This make it easier for the faculty to check a student's academic status at any point of time during the semester.

1.1.4 Role of Educational Analytics

Education Analytics plays an important role in converting academic data into useful information for students and faculty. Instead of calculating the overall semester marks, our system analyses any changes in student performance over different time periods.

By providing clear information about academic performance, the system helps faculty and students to understand difficulties and plan suitable ways to resolve.

1.1.5 Objective of the Digital Class Twin

The main objective of digital class twin is to improve the way how a student's academic performance is monitored for early potential preventions. By combining academic data with risk calculations, our system helps faculty to identify students who require additional support. This eventually provides enough time for faculty to arrange remedial classes, communicate with parents and suggest corrective measures even before students face examinations.

Also our project aims to reduce manual effort, time involved in analysing and improving communication between students, faculty and students. In addition, students can view their own academic status and can understand which reas require improvement.

1.2 Problem Statement and Background

Most existing college ERP systems and student management systems have limitations when it comes to monitoring student performance and identifying the issues of students at an early stage. Based on our requirements, the following four major issues have been identified:

1. **Delayed Performance Processing:** Most college ERP systems calculate attendance performance and total academic results at certain time periods like monthly or end of semester. Due to this, faculty identify students who have shortcomings lately, making it difficult for them to provide sufficient guidance.
2. **Separate Storage:** Information such as attendance, internal marks, laboratory marks and disciplinary reports are maintained separately in a system which makes it difficult to identify academic risks early.

3. **Static Risk Prediction:** Some systems use analytics tools which provide automated predictions in scores or risk percentage without clearly showing the reasons behind it makes it difficult to understand the result.
4. **Absence of Interactive Simulation:** Most ERP's are designed to display only the academic records and attendance details. They don't provide interactive dashboards which makes it easier for the students, faculty or parents to explore the software.

1.3 Objectives

To address these above limitations, our system focuses on the following objectives:

1. To develop an event ingestion to collect regular academic data such as attendance, assignment marks and internal marks.
2. To maintain a digital twin profile for each student to monitor their performance across the semester.
3. To implement a scoring model to calculate the student health score range from 0-100. Based on the score, each student will be be classified into one for these categories: Stable, Warning or Crictical.
4. To develop a What-if simulator that allows students to visualize different academic situation without altering the existing data.

1.4 System Requirements

The software works efficiently on standard college computers and does not require costly GPU hardware. The hardware setup and software dependencies used during testing are presented in Table 1.1.

Table 1.1 Hardware system requirements

Component	Specification
Host Workstation	Standard x64 / ARM64 processor

Processor	Dual-core CPU, 2.0 GHz
RAM	Minimum 4 GB
Storage	Minimum 2 GB disk space

Table 1.2 Software system requirements

Component	Software / Technical Specification
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.10+
Backend Framework	FastAPI, Uvicorn ASGI server, Pydantic v2
Database	SQLite 3
ORM and Database Migration	SQLAlchemy 2.0, Alembic
Authentication and Security	JWT, bcrypt password hashing, Role-Based Access Control (RBAC)
Frontend Framework	React 18
Frontend Development Tool	Vite 7
Routing Library	React Router DOM v6
HTTP Client	Axios
Data Visualization	Chart.js 4
Icon Library	Lucide React

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

To create the foundation for our Digital Class Twin system we looked at twenty papers from 2021 to 2025 in the fields of Educational Data Mining, Learning Analytics and student retention systems. We divided the work among four team members with each person reviewing five papers.

2.2 Educational Data Mining and Supervised Performance Prediction

2.2.1 Associative Classification for Student Performance Prediction

Cagliero et al. (2021) looked at whether rules from quizzes and assignments could predict if students pass a course. Their results showed that early scores can predict results well. For our project this confirmed that tracking Continuous Internal Assessment scores is important. It also showed that we need to track how students do in all courses not just one.

2.2.2 Student Performance Prediction Using Machine Learning Algorithms

Dabhade et al. (2021) compared different machine learning algorithms to predict student performance using different information. This study found that examinations were used for predicting academic performance and gave importance to internal assessment while evaluating students performance. However it mainly focused on prediction rather than continuous support.

2.2.3 Educational Data Mining and Student Performance Prediction

Yagci (2022) applied machine learning techniques to predict student performance. However understanding these models can be difficult. To overcome this our system makes it easier for students to understand.

2.2.4 Student Performance Prediction Using Ensemble Learning

Balcioglu and Artar (2023) used different ensemble learning techniques for predicting student performance. This supported our decision to use random forest algorithm, rather than depending only on static datasets.

2.2.5 Predicting Student Performance in Computer Programming

Su et al. (2022) studied the relationship between assignment submission and exam performance. Based on this observation, we included laboratory performance and practical as important factors in our systems.

2.3 Temporal Learning Analytics, Engagement Logging

2.3.1 Student Engagement Analysis Using Moodle LMS Interaction Logs

Tamada et al. (2022) analysed a huge amount of interaction logs collected from several Leaning Management System (LMS). We applied a similar approach in our system to analyze changes in attendance and laboratory marks.

2.3.2 Student Retention Prediction Using Ensemble Learning

Karalar et al. (2021) used machine learning models to predict student retention learning programmes.. Based on this findings, our system monitors attendance on a regular basis that identifies poor academic performance.

2.3.3 Predicting Student Retention Using Academic Records

Park and Yoo (2021) examined the use of academic data to identify students who may be at risk of dropping out. Based on this, our system includes a What-if simulator that allows students to try different possibilities to improve academic performance.

2.3.4 Student Dropout Prediction Using Learning Analytics

Oqaidi et al. (2022) used university records to build warning systems. They found that early performance is a sign of graduation. We built a system that shows this information on dashboards for students, teachers and parents.

2.3.5 Learning Analytics for Student Persistence Prediction

Segura et al. (2022) studied how early grades and course choices affect whether students stay in school. Segura et al. (2022) Showed that tracking progress over time is important. Our system checks performance to do this.

2.4 Explainable AI (XAI). Dynamic Decision Support in Higher Education

2.4.1 Interpretable Models for Student Engagement Analysis

Katsuragi and Tanaka (2022) developed models to analyze student information about their academic performance. This made us to include an explanation feature in our system.

2.4.2 Explainable Artificial Intelligence for Student Performance Prediction

Nagy and Molontay (2023) demonstrated how it can help identify the factors in an understandable way. Nagy and Molontay (2023) linked model results with advice, in language.

2.4.3 A Unified Approach to Interpreting Model Predictions

Lundberg and Lee (2017) created SHAP, which helps explain model results. We used this to break down each student's risk score into points.

2.4.4 Stability of AI Methods in Predictive Models

Tiukhova et al. (2024) Found that explanation methods can change a lot. This led us to use a model for decisions and machine learning for extra help.

2.4.5 Early Warning. Academic Advising in Higher Education

Plak et al. (2022) Found that early warnings work best when they are linked to help. We included these features such as risk alerts and parent's alerts.

2.5 Research Synthesis and Identified Gaps

Based on the above studies, we identified four major limitations in existing ERP's and prediction systems.

2.5.1 Regular Data Updates

Most existing systems use older data that are updated periodically. Our system updates the student profile whenever new data are entered.

2.5.2 Lack of Data Integration

Students data is stored and maintained in different modules in a software, our system brings every data in one place.

2.5.3 Limited Explainability

Some systems provide risk scores but without explaining the factors behind them.

2.5.4 No What-If Simulation

Existing system only provided few options for academic improvements without exploring different plans to improve.

Table 2.1 Literature review by research area

Research Area	Representative Studies	Relevance to Digital Class Twin
Academic Performance Prediction	Cagliero et al. (2021); Dabhade et al. (2021); Yagci (2022); Su et al. (2022); Balcioglu & Artar (2023)	Demonstrates that formative assessments and quiz telemetry yield strong predictive signals for course outcomes.
Dropout & Failure Forecasting	Karalar et al. (2021); Park & Yoo (2021); Oqaidi et al.	Identifies early-semester attendance lapses and

	(2022); Segura et al. (2022); Katsuragi & Tanaka (2022); Cho et al. (2023); Villar & Robledo Velini de Andrade (2024); Vaarma & Li (2024)	cumulative credits as primary indicators of student attrition.
LMS & Multi-Source Analytics	Tamada et al. (2022); Qin et al. (2023)	Confirms that combining academic records with engagement logs provides a richer student risk profile.
Advisory & Remedial Counseling	Plak et al. (2022); Alalawi et al. (2025)	Emphasizes that predictive alerts are effective only when connected to structured faculty advising and pedagogical support.
Explainable AI & Model Stability	Nagy & Molontay (2023); Tiukhova et al. (2024); Miranda et al. (2024); Villar & Robledo Velini de Andrade (2024)	Establishes that transparent factor attribution (e.g., SHAP, linear weights) is essential for faculty trust and actionable mentoring.

Table 2.2 Summary of twenty representative research papers on student early-warning systems

Sl.	Author & Year	Title & Dataset Focus	Methodology & Models	Key Findings & Identified Gaps
1	**Cagliero et al. (2021)**	Predicting Student Academic Performance by Means of Associative Classification [Course exam telemetry]	Associative classification rules extracted from early-term test and quiz records.	Early test marks correlate strongly with end-semester results. However, the study focused only on single-course data rather than continuous multi-semester student tracking.
2	**Karalar et al. (2021)**	Predicting Students at Risk Using Ensemble Models in Distance Learning [Moodle & Adobe Connect logs]	Ensemble classification (Gradient Boosting, Random Forest, Extra Trees, ANN).	LMS activity logs provide helpful risk indicators for distance learners, but the method lacks integration with physical classroom attendance.
3	**Park & Yoo (2021)**	Early Dropout Prediction in University Online Learning [Institutional	Supervised ML classification across LMS engagement and	Flags dropout risks accurately by mid-semester. However, the pipeline ran as an offline

		online course data]	submission history.	terminal script with no interactive advising or recovery simulation features.
4	**Dabhade et al. (2021)**	Educational Data Mining for Predicting Academic Performance [Demographic and academic dataset]	Comparative evaluation of Decision Trees, Naive Bayes, and SVM classifiers.	Combines academic and demographic variables. However, static batch analysis limits its usefulness for weekly, real-time mentoring interventions.
5	**Tamada et al. (2022)**	Predicting Dropout Risk in Technical Courses Using LMS Logs [1M+ Moodle interaction records]	Random Forest, Decision Tree, Naive Bayes, and K-Means clustering.	Random Forest showed top predictive accuracy on LMS logs, but the study focused strictly on virtual platform clicks rather than in-person lecture attendance.
6	**Yagci (2022)**	Educational Data Mining: Academic Performance Prediction [Undergraduate course dataset]	Supervised machine learning algorithms for categorical grade estimation.	Identifies non-linear patterns between mid-term tests and final grades, but the model remained a black box without factor-by-factor score explanations.
7	**Oqaidi et al. (2022)**	Dropout Prediction Modeling in Higher Education Institutions [University administrative records]	Feature selection filtering combined with multi-class supervised learning.	Confirms the predictive value of first-year grade trends, but the evaluation was entirely retrospective without interactive role-based dashboards.
8	**Segura et al. (2022)**	Student Dropout Prediction: Role of Degree Preferences [Madrid University; 28 institutional variables]	SVM, Decision Tree, ANN, and Logistic Regression with feature analysis.	Student degree choice and starting GPA heavily impact retention. However, testing on a single-institution dataset limits direct cross-campus generalization.
9	**Katsuragi & Tanaka (2022)**	Dropout Prediction by Interpretable Machine Learning Models [High school daily LMS & Slack data]	Interpretable classification using daily communication and activity signals.	Frequent communication drops strongly indicate disengagement, but the approach was limited to online environments and omitted physical classroom roll calls.

10	**Su et al. (2022)**	Exploring Learning Features and Performance in Programming Courses [Quiz logs and code submissions]	Neural networks, Random Forest, SVM, and correlation analysis.	Regular programming exercise submissions correlate strongly with exam scores, though the evaluation was domain-specific to coding coursework.
11	**Plak et al. (2022)**	Early Warning Systems for Effective Student Counseling [VU Amsterdam; two 1st-year cohorts]	Logistic Regression, Additive Models, SVM, and Random Forest in counseling.	Demonstrates that automated early-warning alerts support faculty advising, but the real-world impact depends heavily on teacher adoption and counseling workflows.
12	**Qin et al. (2023)**	Early Warning with Integration of Subjective and Objective Data [Academic, library, and card access logs]	Cost-sensitive neural network combining academic and campus facility usage.	Tracking campus mobility and library check-ins improves risk detection, but multi-sensor campus data collection introduces substantial deployment hurdles.
13	**Cho et al. (2023)**	Dropout Prediction for University Students Using Machine Learning [Multi-year university records]	Ensemble classification with longitudinal feature importance ranking.	Earned credits and GPA trajectories are primary indicators of student dropout, though the model produced only risk labels without recovery planning tools.
14	**Balcioglu & Artar (2023)**	Predicting Academic Performance of Students with Machine Learning [Open University Learning Dataset (OULAD)]	Decision Trees, SVM, Neural Networks, and Stacking Ensembles.	Stacking ensembles achieved high classification accuracy, but the study evaluated static open benchmarks without live event synchronization.
15	**Nagy & Molontay (2023)**	Interpretable Dropout Prediction: XAI for Personalized Intervention [Higher-education cohort data]	Supervised classifiers paired with Explainable AI (SHAP / LIME frameworks).	Model explainability is vital for designing personalized interventions, though their setup did not maintain continuous digital twin trajectories over time.

16	**Villar & Andrade (2024)**	Supervised Algorithms for Student Dropout: A Comparative Study [Institutional academic dataset]	XGBoost, CatBoost, LightGBM, SMOTE/ADASYN resampling, and SHAP.	Gradient boosting with class rebalancing handles minority at-risk cohorts well, though higher model complexity demands greater computational resources.
17	**Vaarma & Li (2024)**	Predicting Student Dropouts: Longitudinal Study in Finnish Higher Ed [Transcripts, demographics, Moodle]	Monthly feature extraction and longitudinal machine learning models.	Predictive feature importance shifts across the semester, highlighting why rolling temporal observation windows are necessary in ongoing student monitoring.
18	**Tiukhova et al. (2024)**	Assessing the Stability of Student Success Models via XAI [Institutional success dataset]	SHAP feature explanations evaluated across temporal cross-validation folds.	Post-hoc explanation methods can vary across academic terms, reinforcing the value of transparent, deterministic scoring formulas for official standing.
19	**Miranda et al. (2024)**	Model-Agnostic Interpretability for Online Performance Prediction [Video-conference learning records]	Random Forest, SVM, Naive Bayes, SMOTE, and Kernel SHAP attributions.	Session length and active participation dominate predictive weight, but a small cohort size (N=361) limits broader institutional conclusions.
20	**Alalawi et al. (2025)**	Learning Analytics Framework Integrating ML and Pedagogical Support	Predictive supervised models integrated into a web-based learning portal.	Connects predictive machine learning models with web-based student support and faculty remediation workflows.

CHAPTER 3

PROPOSED SYSTEM

3.1 System Overview

Our System is designed using six architectural layers, each performing a specific part of the system. The process starts with collecting daily records such as attendance, internal assessment and student disciplinary records. Then the details are entered through API endpoints or uploaded using import options.

Once the data is stored, the engine analyses students academic performance over the last 30 days. It calculates the factors such as attendance percentage and academic performance.

Based on the data, the system calculates health score ranging from 0 to 100 and is classified into three categories: Stable, Warning or Critical.

3.1.1 Layer-by-Layer Architectural Decomposition

Layer 1: Data Ingestion and File Import

This layer is responsible for collecting student information in two ways:

(1) The first method uses REST API endpoints to record continuous data such as attendance, internal marks, and discipline details.

(2) The second method is upload feature using Excel or Csv files.

Layer 2: 30-Day Rolling Performance Analysis

Instead of waiting until the end of the semester, our system continuously analyses their academic performance over the last 30 day period which has four factors: attendance percentage, internal marks, laboratory performance and behaviour.

Layer 3: Academic Health Scoring and Status Tracking

This layer maintains a digital twin for each student and updates continuously. Using the calculated health score, they are classified into three categories:

1. Stable: Students with a health score of 75 or above.
2. Warning: Students with a health score between 50 and 74.
3. Critical: Students with a health score below 50.

Layer 4: Machine Learning Risk and Grade Prediction

With the health score, the system uses two machine learning models to provide insights for students' performance.

1. Random Forest Classification: Used to classify students into High, Medium or Low risk categories.
2. Ridge Regression: Two separate regression models are used in the system. The first models is used to predict expected semester GPA and second is used to predict cumulative CGPA.

Layer 5: Interactive What-If Simulation Sandbox

The What-if simulation module in our systems helps students and faculty provide improvement plans. When a student is identified as at risk, the system provides a simulation without modifying the original data by creating a temporary copy.

This feature provides students and faculty a better way to analyse thor performance.

Layer 6: Multi-User Web and Mobile Interface

Our system provided four different dashboards based on user roles:

(a) Student Dashboard allows students to view their performance streaks, internal marks, attendance and AI based insights

(b) Faculty Dashboard allows to view whole class academic performance, file uploads and alerts about students.

(c) Parent Dashboard allows parents to view their child's academic progress.

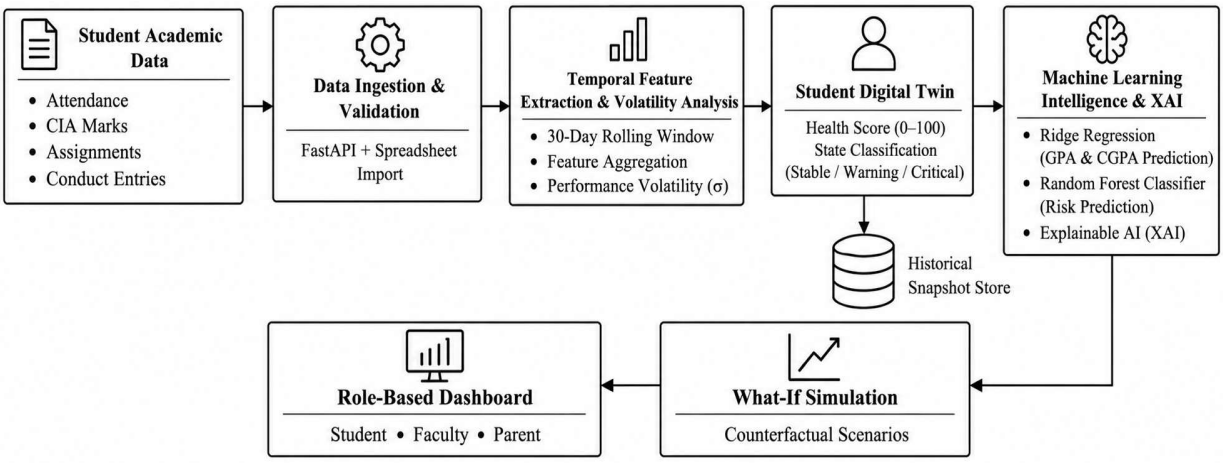


Figure 3.1: Architecture of the Digital Twin System

3.2 Functional Requirements

The system has several functional requirements such as:

Table 3.1 Functional requirements

Req ID	Functional Capability	Prototype Implementation Specification
FR1	**Event Ingestion & Schema Validation**	Accept typed attendance, assessment and conduct events via REST API; validate schema structures using Pydantic.
FR2	**Sliding-Window Feature Extraction**	Derive rolling 30-day attendance rates weighted test. Grade volatility across consecutive assessments.
FR3	**Digital Twin State Machine**	Classify each student profile into Stable, Warning or Critical bands with anti-flicker hysteresis thresholds.
FR4	**Explainable Risk Scoring**	Compute health (0–100) using transparent linear weights (50% academic, 30% attendance 20% conduct) and export factor attributions.
FR5	**Historical Snapshot Archival**	Persist an audit snapshot whenever a student transitions between operational states to prevent data bloat.
FR6	**Multi-Stakeholder Alerting**	Generate scoped notification alerts upon threshold breaches; automatically resolve alerts when standing returns to Stable.

FR7	**Role-Based Web Dashboards**	Deliver specialized web views for Students, Faculty Mentors, Parents and Administrators secured by JWT tokens.
FR8	**Ownership & Privacy Controls**	Enforce relational ownership checks ensuring parents access only mapped children and faculty access assigned sections.
FR9	**What-If Recovery Simulator**	Provide a in-memory shadow twin workspace allowing advisors to project hypothetical score gains without mutating database records.
FR10	**System Health & Audit Logging**	Log authenticated session activity, database transaction durations and administrative configuration modifications.

3.3 Event Ingestion and Student Twin State Model

Regular academic data such as attendance, internal marks and student disciplinary records are passed through HTTP POST requests in JSON format. Table 3.2 shows different health score and action taken.

Table 3.2 Health score threshold configuration

Twin State	Health Score (H)	Risk Index (R)	Triggered Operational Action
Stable	H_composite >= 75.0 (Optimal Academic Health)	R_risk <= 25.0 (Low Risk)	Continuous background monitoring; positive progress feedback.
Warning	50.0 <= H_composite < 75.0 (Moderate Distress)	25.0 < R_risk <= 50.0 (Moderate Risk)	Automated advisor alert; parental advisory notice; remedial counseling queue.
Critical	H_composite < 50.0 (Severe Distress)	R_risk > 50.0 (High Risk)	Urgent faculty action; HOD notification; mandatory academic recovery plan.

3.4 Architecture and Data Flow

The systems backend follows the following approach. Whenever a faculty enters an event, the API validates the data and stores it in the database. Then it continuously updates the students last 30 day data and calculates health score and displays the updated information through the system and alerts are sent.

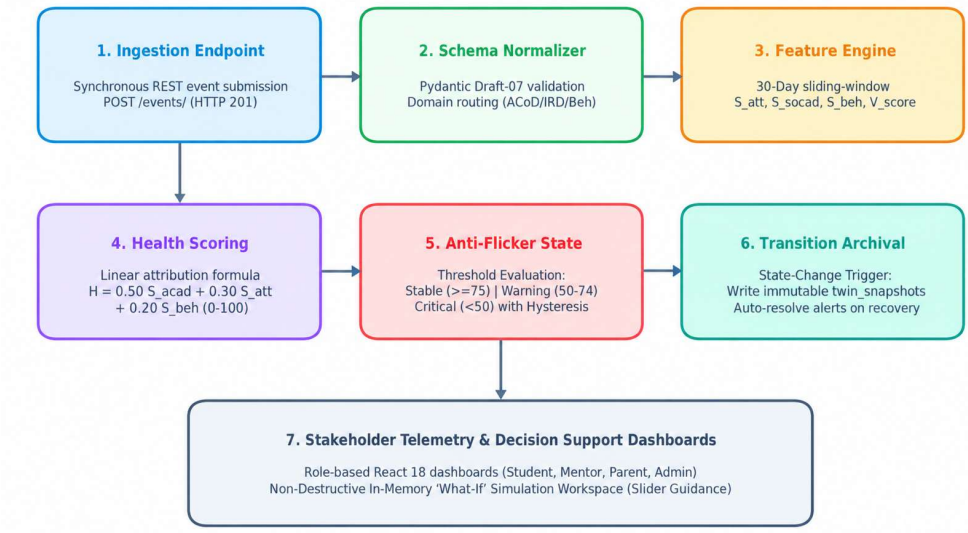


Figure 3.2: Flowchart of the event processing pipeline

3.5 Feature Engineering and Volatility Analysis

Whenever a new event is entered and validated, the backend engine analyses the student performance using the last 30 days and calculates the four metrics.

1. Rolling Attendance Ratio (S_{att})

This represents the student attendance percentage by present sessions and total sessions.

$$S_{\text{att}} = \frac{\text{Present Sessions}}{\text{Total Scheduled Sessions}} \times 100$$

2. Academic Score Average (S_{acad})

This score is the average of Continuous Internal Assessment marks and lab evaluations scaled to a 0–100 range.

3. Behavioral Conduct Index (S_{beh})

This index starts with a score of 100, it increases and decreases when disciplinary log is reported.

$$S_{\text{beh}} = \max(0,\ 100 - 10 \times \text{Sum(Infractions)})$$

4. Academic Volatility Metric (V_{score})

This metric measures the difference in marks between each test and identify any deviation.

$$V_{\text{score}} = \text{Mean}\left(\left|M_j - M_{j-1}\right|\right)$$

Where M_j represents the test score and M_{j-1}represents the previous test mark.

A higher score means more variation in marks between each test. These metrics is useful to identify changes in students academic performance earlier.

```
Algorithm 3.2: 30-Day Sliding-Window Feature Extraction and Volatility
Calculation
Input:  Student identifier student_id, temporal reference t_now, window W = 30
days
Output: Calculated feature vector F = (att_pct, acad_avg, conduct_idx,
volatility, is_provisional)
--------------------------------------------------------------------------------
1:  def extract_student_features(student_id: str, t_now: datetime, window_days:
int = 30):
2:      start_date = t_now - timedelta(days=window_days)
3:      attendance_events =
db.query(Attendance).filter_by(student_id=student_id).since(start_date).all()
4:
5:      # Enforce minimum sample threshold of 3 sessions for reliable percentage
calculation
6:      if len(attendance_events) >= 3:
7:          present_count = sum(1 for e in attendance_events if e.is_present)
8:          att_score = (present_count / len(attendance_events)) * 100.0
9:          att_valid = True
10:     else:
11:         att_score, att_valid = None, False
12:
13:     # Extract normalized exam marks and compute consecutive grade volatility
14:     exam_events =
db.query(Assessment).filter_by(student_id=student_id).since(start_date).all()
15:     if len(exam_events) >= 1:
16:         marks = [(e.score / e.max_score) * 100.0 for e in exam_events]
17:         acad_score = sum(marks) / len(marks)
18:         volatility = (sum(abs(marks[i] - marks[i-1]) for i in range(1,
len(marks))) / (len(marks)-1)) if len(marks) >= 2 else 0.0
19:     else:
20:         acad_score, volatility = None, 0.0
21:
22:     # Derive conduct index with disciplinary penalty deductions
```

```
23:     conduct_events =
db.query(Behavior).filter_by(student_id=student_id).since(start_date).all()
24:     conduct_score = max(0.0, 100.0 - sum(10.0 * b.severity for b in
conduct_events))
25:
26:     return {
27:         'attendance': att_score, 'academic': acad_score,
28:         'conduct': conduct_score, 'volatility': volatility,
29:         'is_provisional': not (att_valid and acad_score is not None)
30:     }
```

3.6 Explainable Student Health Scoring and Twin State Machine

The health score represents a students overall academic score on a range of 0 to 100. Three main factors are used to calculate such as: examination marks(50%), attendance(30%) and behaviour(20%).

The score is calculated as follows:

$$H_{\text{composite}} = 0.50S_{\text{acad}} + 0.30S_{\text{att}} + 0.20S_{\text{beh}}$$

Where:

- S_{acad} = Academic performance score
- S_{att} = Attendance compliance score
- S_{beh} = Behavioral discipline score

The system calculates the Dropout risk index (R_{risk}) by taking the health score and subtracting it from from 100. A high health score means the student is performing well and attending classes regularly. It allows them to provide early support.

Table 3.3 Risk score factors

Attribution Factor	Weight	Input Domain	Diagnostic Meaning
Academic Performance (S_acad)	**0.50 (50%)**	Continuous Internal Assessment (CIA) marks & lab scores [0–100]	Shows how well the student understands the course material and performs in tests.
Attendance Adherence (S_att)	**0.30 (30%)**	30-day rolling present/absent lecture ratio [0–100]	Shows if the student is coming to class regularly and

			participating in labs.
Behavioral Conduct (S_beh)	**0.20 (20%)**	Conduct index with deduction penalties [0–100]	Shows if the student is on time behaves well and takes part in class.

3.7 AI Decision Support, Alerting Engine and Intervention Simulation

Whenever a students performance changes to warning or critical, the system automatically sends an alert to the faculty profile. In order to prevent repeated notification, it does not send the alert unless until the issue is still active.

The system provides a 3 point buffer to prevent sending repeated alerts when health score is near to the threshold.

3.8 Role-Based Dashboards, Data Privacy and Technology Stack

Our system used JWT with HMAC-SHA256 for secure user access and identity based on their roles. The system has 3 types of users: Students, Faculty and Parents. Table 3.4 provides the endpoints used in the platform.

Table 3.4 REST API endpoints

HTTP Verb	Route Path	Operational Responsibility
POST	**/auth/login**	Authenticates user credentials, validates assigned role, and issues signed JWT bearer token.
POST	**/events/**	Validates and ingests typed academic events synchronously, executing in-line twin feature derivation and state evaluation (HTTP 201).
GET	**/twins/{student_id}**	Retrieves live digital twin metrics, health score, risk band, and latest snapshot telemetry.
GET	**/twins/{student_id}/history**	Fetches chronological snapshot timeline for longitudinal trajectory rendering.
GET	**/insights/{student_id}**	Returns explainable factor attributions and rule-based diagnostic guidance.
GET	**/alerts**	Lists active, role-scoped academic risk

		notifications classified by severity.
POST	**/dashboard/whatif**	Calculates in-memory shadow twin projection under hypothetical attendance and exam parameters.
POST	**/api/ml/predict-risk**	Runs Random Forest classification with dynamic tree feature attribution (HTTP 200).
POST	**/api/ml/predict-cgpa**	Executes Ridge Regression continuous predictor for projected semester SGPA and cumulative CGPA.
POST	**/api/ml/what-if-simulate**	Simulates hypothetical score gains through trained ML models without database mutation.
POST	**/api/upload/preview**	Fuzzy-matches CSV/Excel column headers and returns schema mapping suggestions.
POST	**/api/upload/commit**	Batch ingests tabular student records, synchronizes digital twins, and triggers ML evaluation.

CHAPTER 4

SYSTEM TESTING

4.1 Test Approach

Our system was tested using automated unit tests, REST API integration tests and performance checks. The unit tests helps us to test whether all the calculations are working, score changes also measured and the defined limits. The integration tests verify whether API request are integrated correctly with the database.

4.2 Functional and Integration Scenarios

We tested scenarios as mentioned in the table 4.1 from T1 to T9 using pytest and FastAPI TestClient and results obtained during tests.

Table 4.1 Functional components test scenarios

Test ID	Evaluation Scenario	Expected System Response	Status
T1	Baseline Normal Events	Twin stays in state; health score is 75 or higher; no adverse alerts are sent.	**PASSED**
T2	Consecutive Absenteeism	Attendance falls below 60 percent; twin moves to Warning state; advisor alert is raised.	**PASSED**
T3	Exam Score Degradation	CIA marks fall below 40 percent; composite health drops into level which is less than 50.	**PASSED**
T4	Anti-Flicker Hysteresis	Scores that swing around the boundary do not cause multiple alerts to be created.	**PASSED**
T5	Snapshot Generation	Twin Snapshot is saved when a valid state change happens; the timeline is updated	**PASSED**
T6	Parent Scope Isolation	A parent who tries to access a student receives HTTP 403 Forbidden.	**PASSED**
T7	What-If Shadow Projection	Adding a 20 percent attendance gives a projected score but does not change the database.	**PASSED**

T8	Malformed Event Schema	An invalid JSON payload is rejected with HTTP 422 Unprocessable Entity.	**PASSED**
T9	Password Security (Bcrypt)	Salting and hashing are used to verify credentials and timing attacks are prevented.	**PASSED**

4.3 Non-Functional Verification Plan

The system is evaluated by checking its reliability, processing time and security of data. Table 4.2 shows the benchmark criteria during prototype testing.

Table 4.2 Non Functional components test scenarios

Quality Attribute	Design Acceptance Target	Observed Prototype Metric
API Response Latency	< 200 ms per HTTP transaction.	The automated FastAPI test runner shows a latency of 42 milliseconds end-to-end.
Event Ingestion Throughput	>= 100 events/sec (Enterprise target).	The prototype processes 38.3 events per second while using single-threaded SQLite write locks, which meets the roll-call needs of 60 students in 1.57 seconds.
Security & Privacy	Zero plain-text password or JWT token leaks.	Cryptographic hash verification is done with direct bcrypt; tokens are signed with HS256; role isolation is enforced.
Explainability Coverage	100% of risk scores yield factor breakdowns.	We verified all test cases with academic, attendance and conduct attributions.
Client Render Performance	Vite build < 5s; responsive UI rendering.	The production Vite build finished in 3.95 seconds. We verified that Chart.js telemetry renders correctly.

4.4 Evaluation Limits

While this testing shows that the architecture is correct that database transactions stay consistent and that the interface responds well it is not a substitute for testing with real student populations, over semesters. We plan studies using historical college records before launching to the whole campus.

CHAPTER 5

RESULT AND DISCUSSION

5.1 Operational Verification and Implementation Evidence

We tested the platform using student profiles that represent students to make sure that the digital twin changes correctly when attendance and exam marks change. Student S001, who had than 90% attendance and an 85% average in internal tests stayed in the Stable tier with a health score of 87.2 and no alerts. On the hand Student S002, whose attendance fell to 58% and test marks went below 45% moved from Warning to Critical with a health score of 44.8.

Table 5.1 Subsystem components and status

Subsystem Component	**Operational Status & Capability**
FastAPI Backend Server	Deployed & Live on Render Cloud
Database Models & Seeding	Relational ORM entities initialized; realistic student profiles seeded.
Twin State Pipeline	Operational on Live Backend; Continuously tracks rolling 30-day telemetry, computes dynamic state transitions (stable, warning, critical), and generates immutable TwinSnapshot audit logs on state transitions.
What if Simulation	Active in Live Dashboard; In-memory simulation workspace allowing advisors and students to model projected recovery scores in real time without persisting fake data
Role & Access Control	Parent-Child and Faculty-Class maps strictly enforce record privacy.

5.1.1 Role-Based Access Control and Authentication Portal

The system has three main dashboards such as student, parent and faculty modules with authenticated role based access control (RBAC).

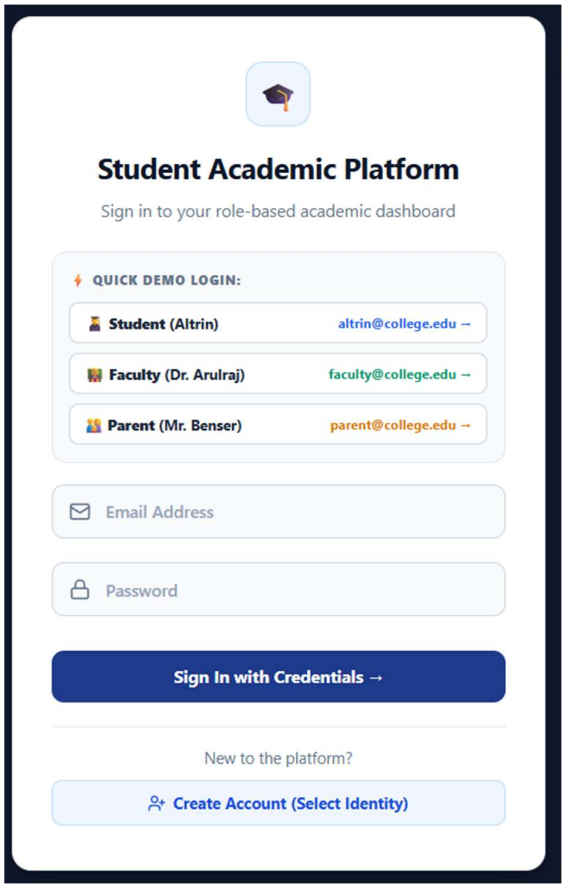


Figure 5.1: User login screen

5.1.2 Primary Stakeholder Dashboards

Once signed in users go directly to a dashboard that matches their roles. Student Academic Dashboard (Figure 5.2): The student dashboard shows a look at academic health. It shows a predicted semester CGPA of 7.15 out of 10.0 79% attendance with a badge showing 4% above the 75% university limit and a risk index of 12%. Interactive cards let students go to exam schedules, attendance records and a What-If recovery tool.

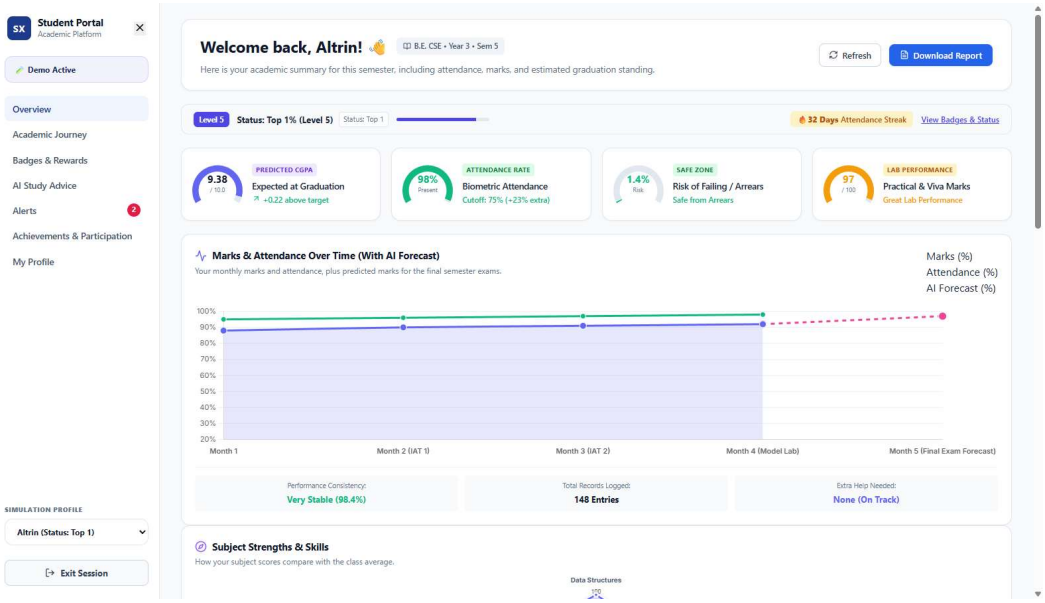


Figure 5.2: Student dashboard

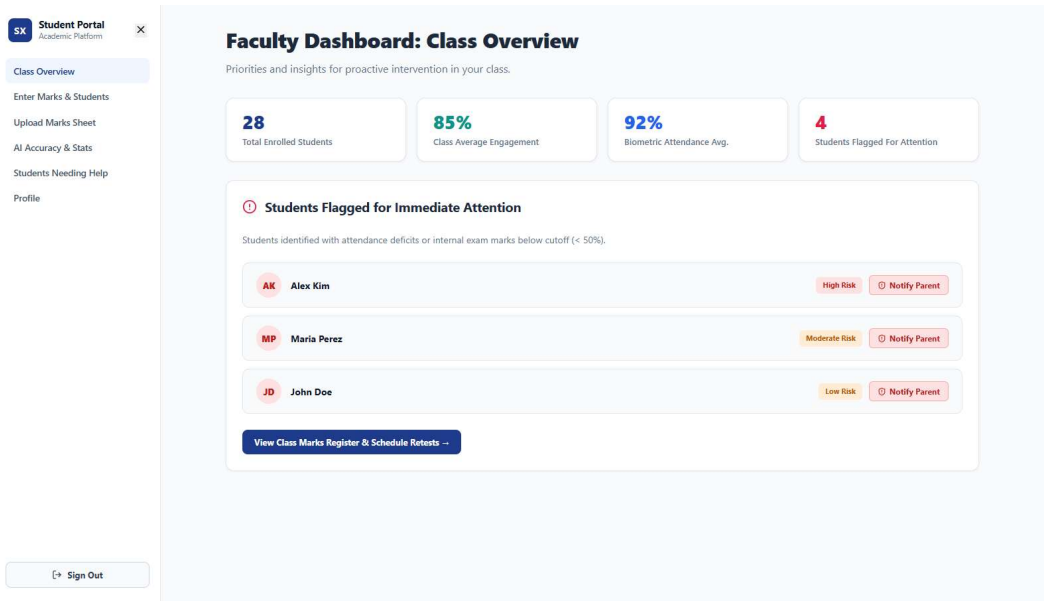


Figure 5.3: Faculty dashboard

Parent Portal Overview (Figure 5.4): The parent screen shows the childs attendance compared to the 75% university rule, recent test scores and notes from the mentor.

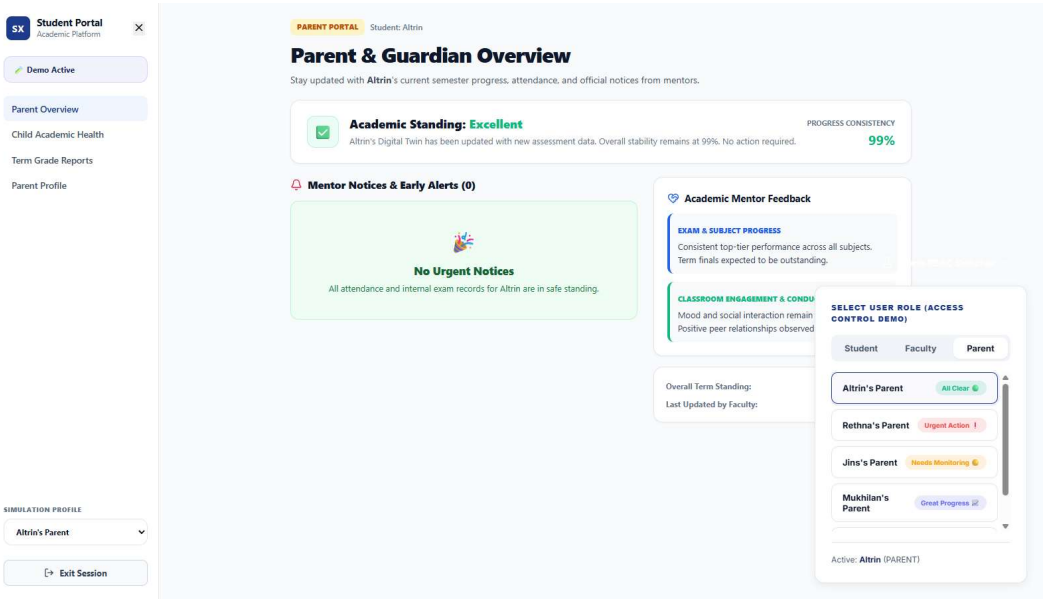


Figure 5.4: Parent dashboard

5.1.3 Core Functional Modules and Decision Support Tools

Apart from the dashboards, the system has several other features such as student tracking, Explainable Ai insights, Multi format uploading and identified students who need support.

The timeline displays important events across the semester such as examinations and important dates. This helps students to understand their academic status and can understand areas where they need to improve. Figure 5.5 shows the digital profile of a student.

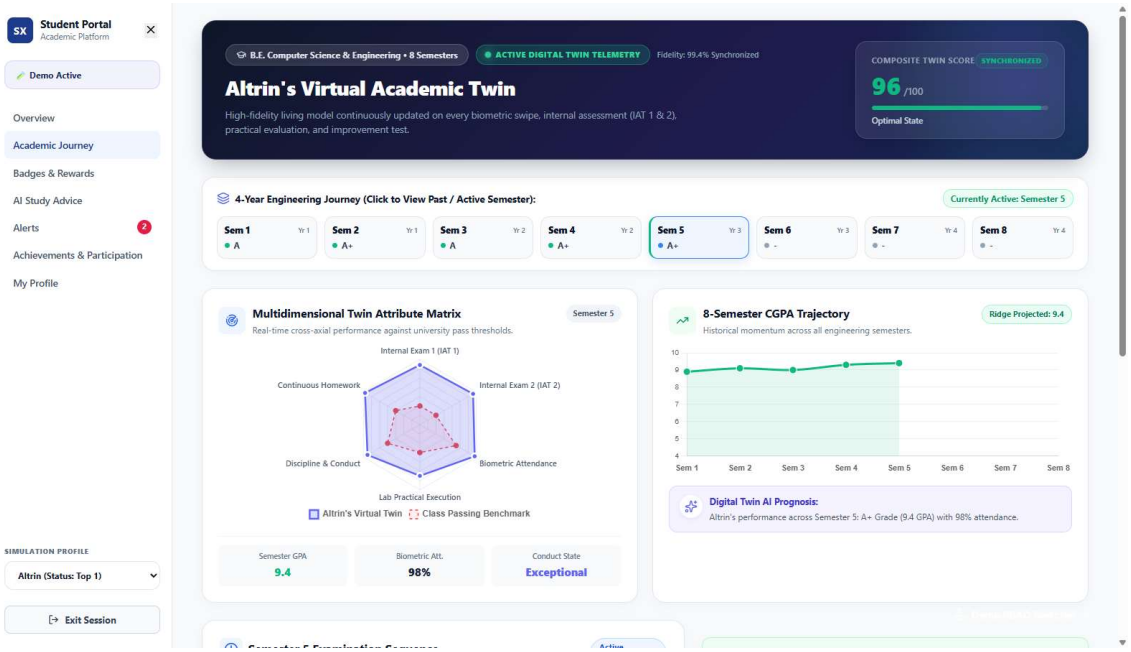


Figure 5.5: Student performance charts

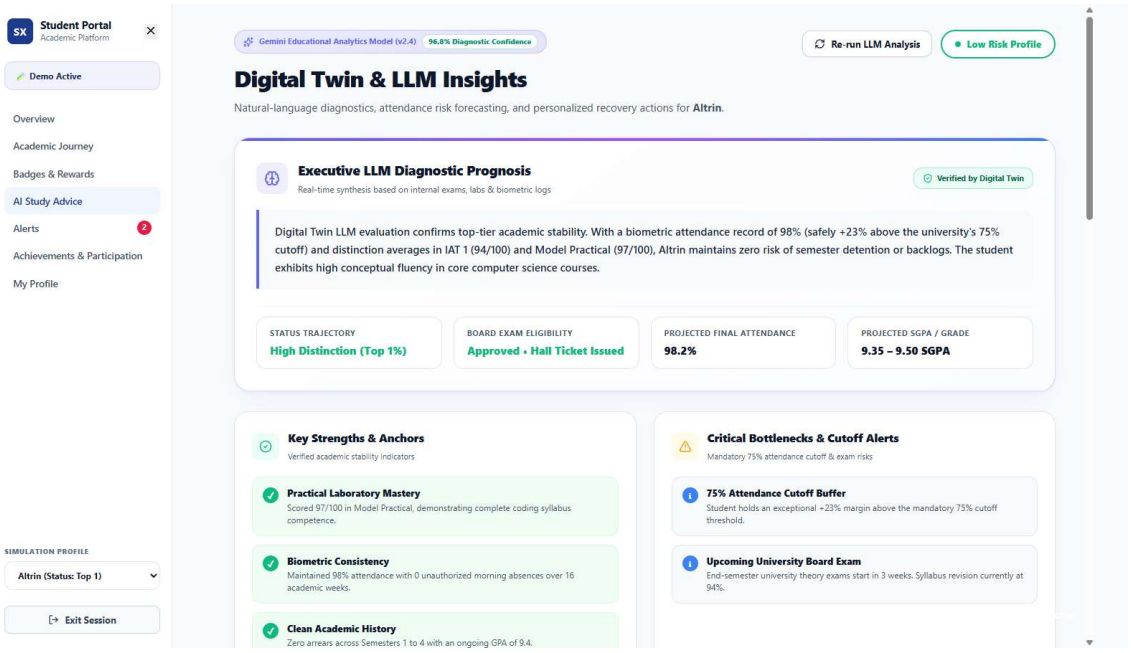


Figure 5.6: Explainable AI insights

Dynamic Multi-Format Upload (Figure 5.7): Figure 5.7 displays how faculty can upload student data using Excel or Csv. The system detects the needed columns and validates the data to make sure the information is correct.

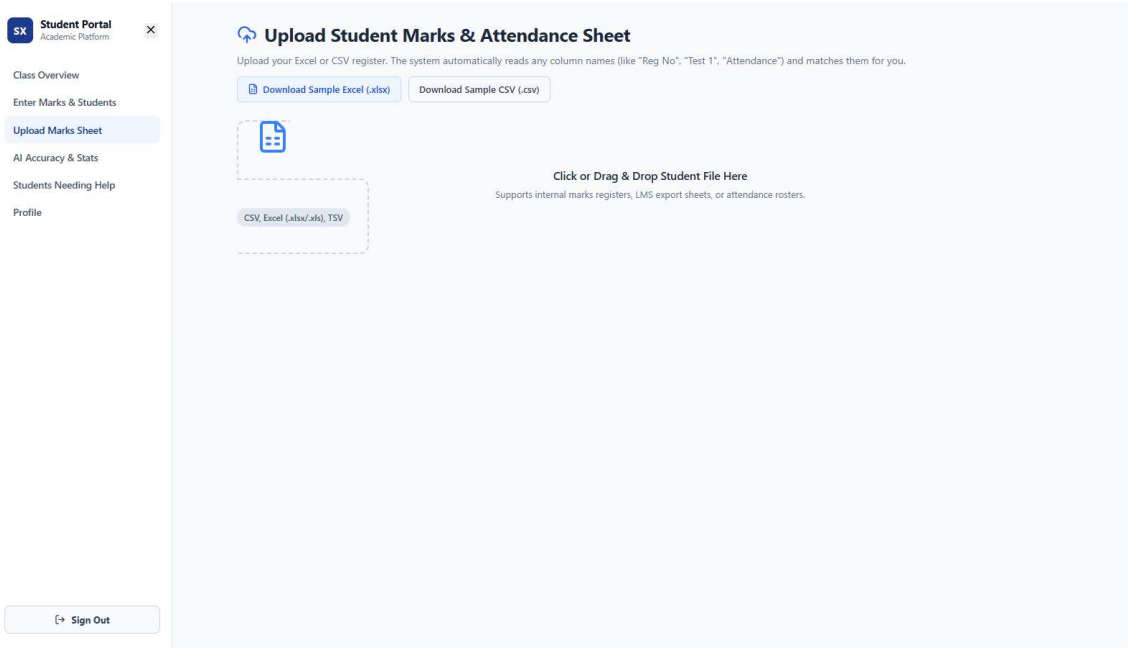


Figure 5.7: File upload using Excel or Csv

Figure 5.8 shows faculty dashboard where they can view students who are identified under warning or critical stage. Based on these details, faculty can send alert to student, and inform their parents and use what if tool to improve their performance in different ways.

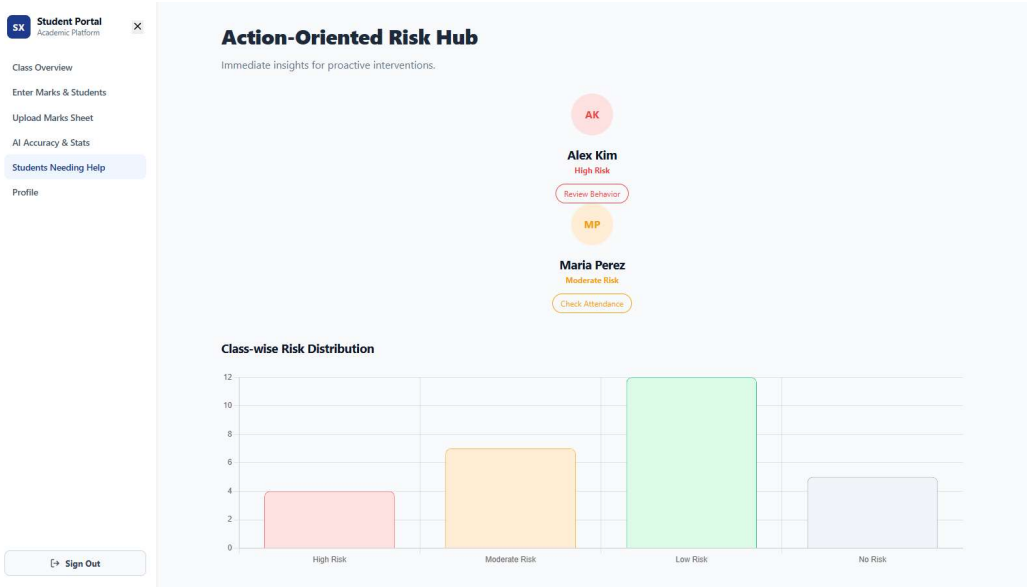


Figure 5.8: Faculty dashboard and students alerts

5.2 Empirical Benchmarks and Latency Analysis

The laptop runs Python 3.13 and SQLite across 100 ingestion runs (Figure 5.9): (1) Event Ingestion Latency: Processing an attendance or score event and updating the 30-day feature vector averaged 26.12 ms. The time ranged between 18.41 ms and 38.22 ms. This is below our 200 ms target. (2) Student State Query Time: Loading a digital twin profile took an average of 2.24 ms. (3) Event Throughput: With SQLites single-file write lock the system handled 38.3 events per second. It processed a roll-call of 60 students in 1.57 seconds.

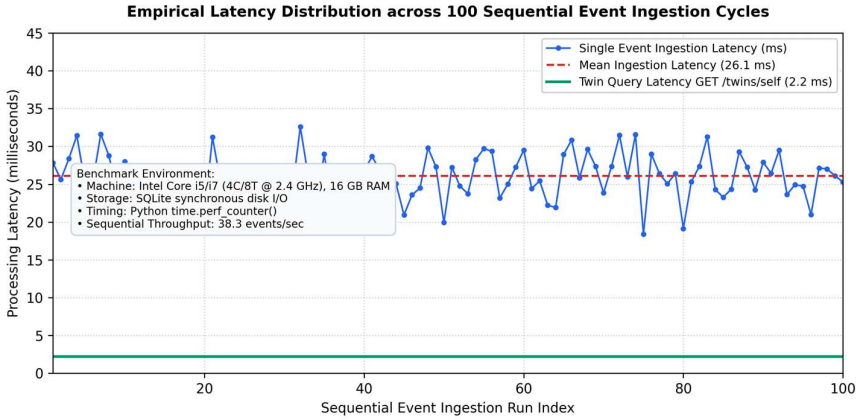


Figure 5.9: System response time across 100 test runs

5.3 Discussion of Architectural and Advisory Findings

The system was tested under different conditions to overcome three advantages of the system. Firstly, a 30 day window is used to help identify differences in attendance and academic performance even before the end of semester. Secondly, Explainable Ai used to understand why a student was identified as risk alert. Finally a What if simulator to perform future academic goals.

5.4 Practical System Limitations and Engineering Constraints

Two practical engineering choices shaped our work. First the transparent baseline formula that has 50% academics, 30% attendance and 20% conduct

gives a to-understand score, for mentoring sessions. We used Random Forest and Ridge Regression models to handle non linear patterns in the data. SQLite was used for testing initially. For college deployment, we plan to integrate to PostgreSQL to improve database performance.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

This project focused on creating a working prototype of an AI-Driven Academic Risk and Dropout Early-Warning System that is built around the Digital Class Twin concept. We designed, implemented and tested a system that uses event ingestion to collect data. It processes this data using a 30-day sliding window for analytics. The system calculates a health score for each student. It also applies machine learning models to forecast academic risk. The results are displayed through role based dashboards. The main aim is to identify students who need support before the end of the semester period.

The system was developed using Python FastApi for the backend, SQLite is used for data storage and React 18 is used for user interface. However the prototype performed well during testing, even further validation is required before deploying. Thus allowing us to fine tune the model factors and improve accuracy.

6.2 Future Enhancement

Our system improvements are divided into three phases as mentioned in table 6.1.

Table 6.1 Implementation and Deployment plan

Deployment Phase	Planned Functional Enhancements	Technical Validation Focus
Phase 1: Core Build (Completed)	Event-driven ingestion pipeline, rule-based attribution engine, SQLite storage, role-based React dashboards in-memory What-If simulator.	Unit testing, integration suites, test harness verification via

		verify_digital_twin.py.
Phase 2: ML Model Training	Train supervised XGBoost / Random Forest models, on multi-semester institutional datasets; calibrate weights.	Cross-validation ROC-AUC precision-recall optimization, SHAP tree explainers.
Phase 3: Production Rollout	PostgreSQL database migration Celery/Redis task queues SMS/WhatsApp parent alerts, mobile app release	Enterprise load testing, security audits, multi-college multi-tenant deployment.

APPENDICES

Appendix 1: Module Responsibility Split and Milestones

The project work was divided among the four team members. The tasks assigned for each members are presented in the table below.

Table A.1 Member wise responsibility

Team Member	Assigned Functional Subsystem	Core Deliverables
Altrin Benser. A (962223205010)	**System Architecture & Integration**	Project coordination, architecture design, repository maintenance, end-to-end pipeline integration.
Mukhilan. M (962223205040)	**Backend Pipeline & Database**	FastAPI architecture, SQLAlchemy ORM models, database migrations, demo data seed scripts.
John Jins. J (962223205032)	**Digital Twin Engine & Alerting**	Twin state machine, snapshot timeline logic, multi-tier alert engine, notification routing.
Jerish Rethna. J (962223205029)	**ML Engine, Features & UI**	Feature derivation, volatility metrics, React client dashboard UI, What-If simulator, verification scripts.

Appendix 2: Core Data Contracts and Schemas

The formal JSON schema defining synchronous academic event ingestion is defined as follows:

AcademicEventIngestionSchema (JSON Schema v7)

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "AcademicEventIngestionSchema",
  "type": "object",
  "required": ["child_id", "event_type", "payload"],
  "properties": {
    "child_id": { "type": "string", "description": "Student institutional ID" },
    "event_type": { "type": "string", "enum": ["attendance", "academic", "behavior"] },
    "payload": {
      "type": "object",
```

```
    "properties": {
      "present": { "type": "boolean" },
      "score": { "type": "number", "minimum": 0, "maximum": 100 },
      "severity": { "type": "integer", "minimum": -5, "maximum": 5 }
    }
   }
  }
}
```

Appendix 3: Review Demonstration Scenarios

Table B.1 outlines the end-to-end user demonstration scenarios evaluated during review sessions.

Table B.1 End-to-end review demonstration scenarios

Scenario ID	User Action / Test Input	Observed UI & System Response
S1: Baseline Login	Baseline Login Faculty logs into the dashboard using -configured demonstration authentication.	Role is recognized; the user is redirected to the Class Overview page that displays the student roster.
S2: Event Ingestion	Event Ingestion Teacher submits three absences for a student.	The event is processed, the attendance feature decreases and the twin status shifts from Stable to Warning.
S3: Alert Generation	Alert Generation Student risk drops into the Warning threshold.	An alert banner appears on both the Faculty and Parent dashboards complete with tags.
S4: What-If Simulation	What-If Simulation Counselor adjusts a test score to 85% on the student profile.	The shadow twin projects a recovery, to the band but no changes are saved.
S5: Parent View Check	Parent View Check Parent logs into the portal using the mapped credentials.	The child twin is. Privacy is verified as only the assigned child is visible.

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